10/29/2026

MID-TERM EXAM

SUBMITTED TO :

USMAN AKRAM

SUBMITTED BY:

Muhammad Saad Lodhi

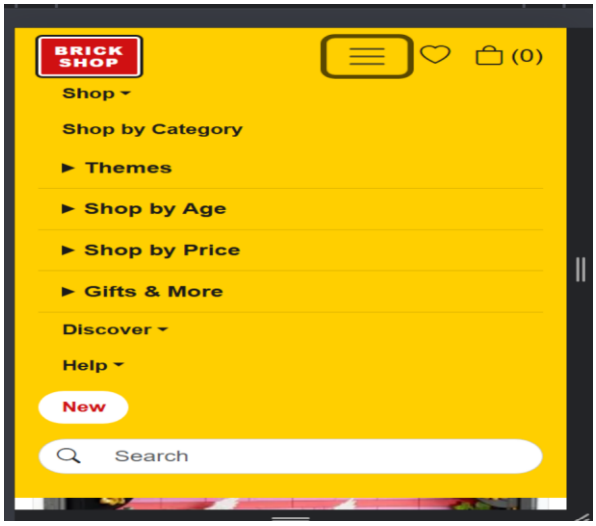
REGISTRATION NO:

FA24-BCS-077

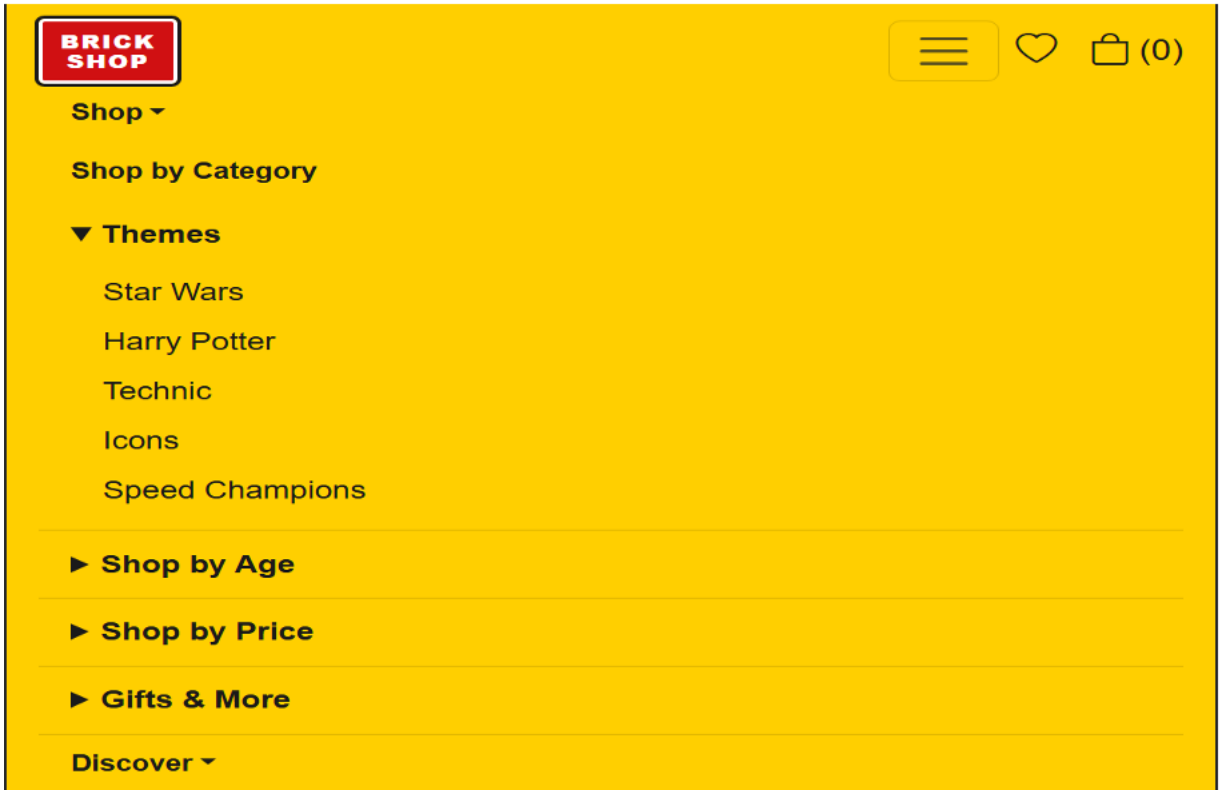
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Task 1 – Mega-menu in the navbar

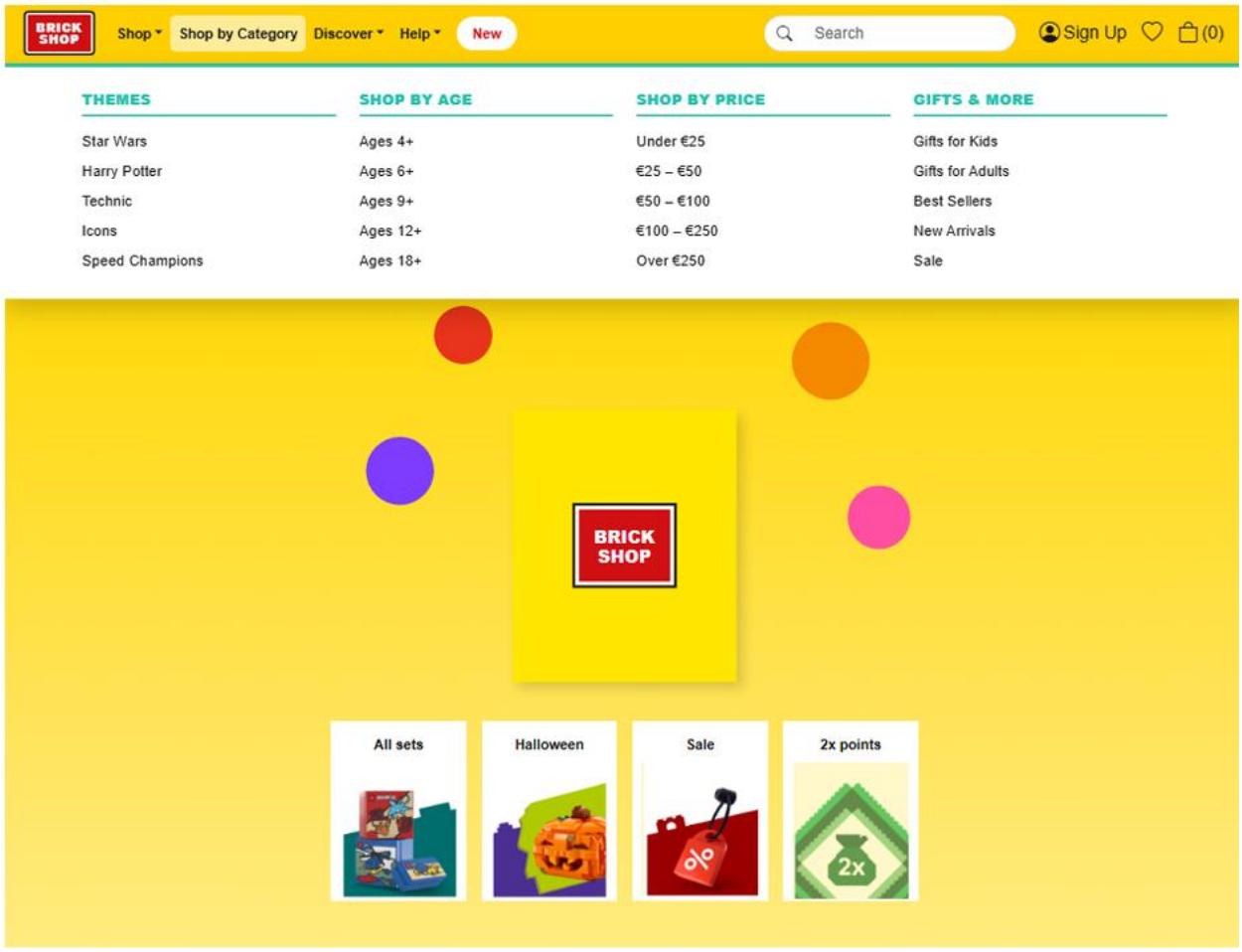
375px (hamburger menu, accordion closed)



604px (BP):



1280px:


BRICK SHOP
Shop
Shop by Category
Discover
Help
New
Search
Sign Up
(0)
THEMES
Star Wars
Harry Potter
Technic
Icons
Speed Champions
SHOP BY AGE
Ages 4+
Ages 6+
Ages 9+
Ages 12+
Ages 18+
SHOP BY PRICE
Under €25
€25 – €50
€50 – €100
€100 – €250
Over €250
GIFTS & MORE
Gifts for Kids
Gifts for Adults
Best Sellers
New Arrivals
Sale
BRICK SHOP
All sets
Halloween
Sale
2x points
%
2x

Task 1 – Code

index.html (and product.html, checkout.html) – changed navbar lines

```
<button class="navbar-toggler ms-auto me-2" type="button" data-bs-toggle="collapse"
    data-bs-target="#mainNav" aria-controls="mainNav" aria-expanded="false" aria-label="Toggle
    navigation">
<div class="header-actions d-flex align-items-center gap-3">
<div class="collapse navbar-collapse" id="mainNav">
```

index.html – mega-menu markup (desktop panel + mobile accordion)

```
<!-- ===== TASK 1: MEGA MENU ===== -->
<li class="nav-item mega-item">
  <a class="nav-link mega-trigger" href="#" aria-haspopup="true">Shop by Category</a>

  <!-- Desktop version (shown from BP and wider): K = 4 columns -->
  <div class="mega-panel">
    <div class="container">
       <div class="mega-grid">
         <div class="mega-col">
          <h3 class="mega-heading" id="mc-themes">Themes</h3>
          <ul aria-labelledby="mc-themes">
            <li><a href="#">Star Wars</a></li>
            <li><a href="#">Harry Potter</a></li>
            <li><a href="#">Technic</a></li>
            <li><a href="#">Icons</a></li>
            <li><a href="#">Speed Champions</a></li>
          </ul>
        </div>
        <div class="mega-col">
          <h3 class="mega-heading" id="mc-age">Shop by Age</h3>
          <ul aria-labelledby="mc-age">
            <li><a href="#">Ages 4+</a></li>
            <li><a href="#">Ages 6+</a></li>
            <li><a href="#">Ages 9+</a></li>
            <li><a href="#">Ages 12+</a></li>
            <li><a href="#">Ages 18+</a></li>
          </ul>
        </div>
        <div class="mega-col">
          <h3 class="mega-heading" id="mc-price">Shop by Price</h3>
          <ul aria-labelledby="mc-price">
            <li><a href="#">Under €25</a></li>
            <li><a href="#">€25 – €50</a></li>
            <li><a href="#">€50 – €100</a></li>
            <li><a href="#">€100 – €250</a></li>
            <li><a href="#">Over €250</a></li>
          </ul>
        </div>
        <div class="mega-col">
          <h3 class="mega-heading" id="mc-gifts">Gifts &amp;amp; More</h3>
          <ul aria-labelledby="mc-gifts">
            <li><a href="#">Gifts for Kids</a></li>
            <li><a href="#">Gifts for Adults</a></li>
            <li><a href="#">Best Sellers</a></li>
            <li><a href="#">New Arrivals</a></li>
            <li><a href="#">Sale</a></li>
          </ul>
        </div>
      </div>
    </div>
  </div>

  <!-- Mobile version (below BP): same K groups as an accordion -->
  <div class="mega-accordion">
    <details>
      <summary>Themes</summary>
      <ul>
        <li><a href="#">Star Wars</a></li>
        <li><a href="#">Harry Potter</a></li>
        <li><a href="#">Technic</a></li>
        <li><a href="#">Icons</a></li>
        <li><a href="#">Speed Champions</a></li>
      </ul>
    </details>
```

```
    <details>
      <summary>Shop by Age</summary>
      <ul>
        <li><a href="#">Ages 4+</a></li>
        <li><a href="#">Ages 6+</a></li>
        <li><a href="#">Ages 9+</a></li>
        <li><a href="#">Ages 12+</a></li>
        <li><a href="#">Ages 18+</a></li>
      </ul>
    </details>
    <details>
      <summary>Shop by Price</summary>
      <ul>
        <li><a href="#">Under €25</a></li>
        <li><a href="#">€25 – €50</a></li>
        <li><a href="#">€50 – €100</a></li>
        <li><a href="#">€100 – €250</a></li>
        <li><a href="#">Over €250</a></li>
      </ul>
    </details>
    <details>
      <summary>Gifts &amp;amp; More</summary>
      <ul>
        <li><a href="#">Gifts for Kids</a></li>
        <li><a href="#">Gifts for Adults</a></li>
        <li><a href="#">Best Sellers</a></li>
        <li><a href="#">New Arrivals</a></li>
        <li><a href="#">Sale</a></li>
      </ul>
    </details>
  </div>
</li>
<!-- ===== END TASK 1 ===== -->
```

style.css – Task 1 CSS

```
/* ======================================================================
   TASK 1 - MEGA MENU (N=4 -> BP=604px, H=28, K=4, cart limit=2)
   Change these 3 things if your N is different:
     1) --accent hue       (H)
     2) @media min-width   (BP)
     3) repeat(4, 1fr)     (K)  + the number of .mega-col / <details> in HTML
   ====================================================================== */
:root { --accent: hsl(28, 70%, 45%); }    /* hsl(H, 70%, 45%) */

/* ---------- DEFAULT = BELOW BP (mobile first)---------- */
.mega-panel { display: none; }                  /* desktop panel hidden */
.mega-trigger { pointer-events: none; }         /* just a label in hamburger menu */

.mega-accordion details { border-bottom: 1px solid rgba(0,0,0,.15); }
.mega-accordion summary {
  cursor: pointer; font-weight: 600; padding: .6rem 1rem;
}
.mega-accordion summary:hover,
.mega-accordion summary:focus-visible { color: var(--accent); }
.mega-accordion ul { list-style: none; margin: 0; padding: 0 1rem .6rem 2rem; }
.mega-accordion a { display: block; padding: .25rem 0; color: var(--lego-black); text-decoration: none;
    }
.mega-accordion a:hover, .mega-accordion a:focus-visible { color: var(--accent); text-decoration:
    underline; }

/* ---------- FROM BP AND WIDER----------*/
@media (min-width: 604px) {

  /* 1) Make the navbar expand at BP instead of Bootstrap's 992px */
  .site-header .navbar-toggler { display: none; }
  .site-header .navbar-collapse { display: flex !important; flex-basis: auto; order: 2; }
  .site-header .navbar-nav { flex-direction: row; }
  .site-header .navbar-nav .dropdown-menu { position: absolute; }
  .site-header .header-actions { order: 3; }
  .site-header .nav-link { padding: .5rem .7rem; }
  .site-header .search-form { margin: 0 1rem 0 auto; min-width: 160px; }

  /* 2) Hide the accordion, show the mega panel logic */
  .mega-accordion { display: none; }
  .mega-trigger { pointer-events: auto; }

  /* The panel is positioned against .navbar (position: relative in Bootstrap),
```

```
      so it is full width, sits right under the navbar and takes NO space in the layout */
  .mega-panel {
    display: block;
    position: absolute; top: 100%; left: 0; right: 0;
    z-index: 1030;
    background: #fff;
    border-top: 4px solid var(--accent);
    box-shadow: 0 12px 24px rgba(0,0,0,.18);
    padding: 1.5rem 0;

    visibility: hidden; opacity: 0;
    transition: opacity .15s, visibility .15s;
  }

  /* Invisible bridge: fills the small gap between the menu item and the panel
     so the panel does not close while the mouse travels down */
  .mega-panel::before {
    content: ""; position: absolute; left: 0; right: 0; top: -1.5rem; height: 1.5rem;
  }

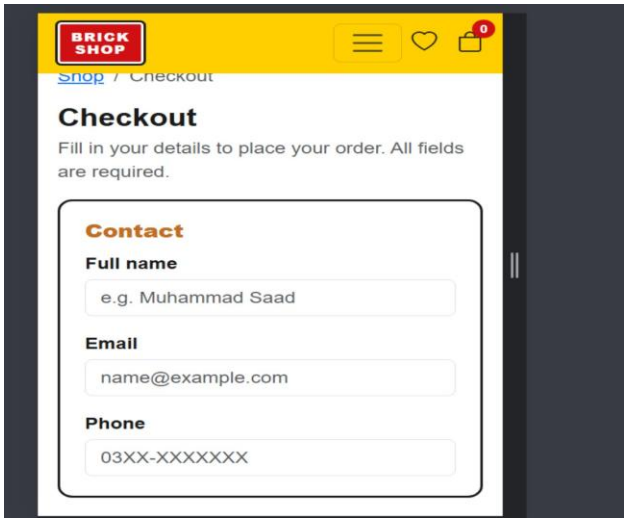
  /* OPEN on hover OR when ANY child has keyboard focus (Tab) */
  .mega-item:hover .mega-panel,
  .mega-item:focus-within .mega-panel {
    visibility: visible; opacity: 1;
  }
  .mega-item:hover > .mega-trigger,
  .mega-item:focus-within > .mega-trigger { background: rgba(255,255,255,.6); border-radius: 6px; }

  /* K columns */
  .mega-grid { display: grid; grid-template-columns: repeat(4, 1fr); gap: 1.5rem; }

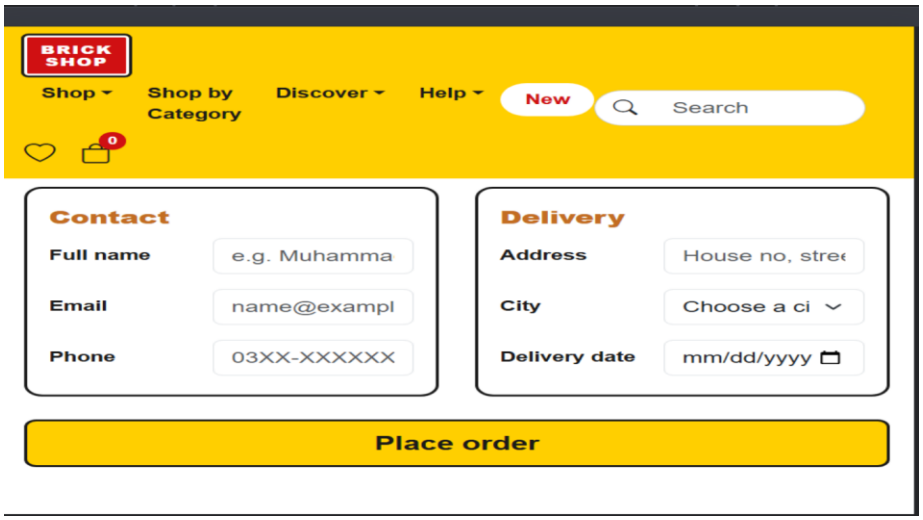
  .mega-heading {
    font-size: .95rem; font-weight: 800; text-transform: uppercase;
    color: var(--accent); border-bottom: 2px solid var(--accent);
    padding-bottom: .4rem; margin-bottom: .6rem;
  }
  .mega-col ul { list-style: none; margin: 0; padding: 0; }
  .mega-col a { display: block; padding: .25rem 0; color: var(--lego-black); text-decoration: none;
      font-size: .95rem; }
  .mega-col a:hover, .mega-col a:focus-visible { color: var(--accent); text-decoration: underline; }
}
```

Task 2 – Checkout form (HTML5 validation only)

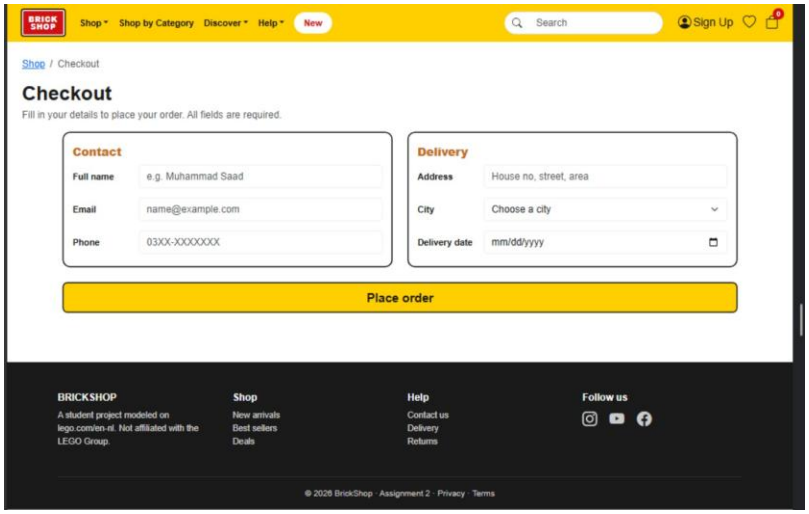
375px (one column, labels above inputs)



604px (BP: fieldsets side by side)



1280px


BRICK SHOP
Shop
Shop by Category
Discover
Help
New
Search
Sign Up
Shop / Checkout
Checkout
Fill in your details to place your order. All fields are required.
Contact
Full name
e.g. Muhammad Saad
Email
name@example.com
Phone
03XX-XXXXXXX
Delivery
Address
House no, street, area
City
Choose a city
Delivery date
mm/dd/yyyy
Place order
BRICKSHOP
A student project modeled on lego.com/en-nl. Not affiliated with the LEGO Group.
Shop
New arrivals
Best sellers
Deals
Help
Contact us
Delivery
Returns
Follow us
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Task 2 – Code

checkout.html – form

```
<!-- ===== TASK 2: HTML5 validation only (no JavaScript) ===== -->
<form class="checkout-form" action="#" method="get">

  <fieldset class="checkout-fieldset">
    <legend>Contact</legend>

    <div class="field">
      <label for="fullname" class="form-label">Full name</label>
      <input id="fullname" name="fullname" type="text" class="form-control"
             required minlength="3" autocomplete="name"
             placeholder="e.g. Muhammad Saad" aria-describedby="fullname-err">
      <span class="error-msg" id="fullname-err">Enter your full name (at least 3 characters).</span>
    </div>

    <div class="field">
      <label for="email" class="form-label">Email</label>
      <input id="email" name="email" type="email" class="form-control"
             required autocomplete="email"
             placeholder="name@example.com" aria-describedby="email-err">
      <span class="error-msg" id="email-err">Enter a valid email address.</span>
    </div>

    <div class="field">
      <label for="phone" class="form-label">Phone</label>
      <input id="phone" name="phone" type="tel" class="form-control"
             required pattern="03[0-9]{2}-[0-9]{7}" inputmode="tel"
             placeholder="03XX-XXXXXXX" title="Format: 03XX-XXXXXXX"
             aria-describedby="phone-err">
      <span class="error-msg" id="phone-err">Use the format 03XX-XXXXXXX.</span>
    </div>
  </fieldset>

  <fieldset class="checkout-fieldset">
    <legend>Delivery</legend>

    <div class="field">
      <label for="address" class="form-label">Address</label>
      <input id="address" name="address" type="text" class="form-control"
             required minlength="5" autocomplete="street-address"
             placeholder="House no, street, area" aria-describedby="address-err">
      <span class="error-msg" id="address-err">Enter your address (at least 5 characters).</span>
    </div>

    <div class="field">
      <label for="city" class="form-label">City</label>
      <select id="city" name="city" class="form-select" required aria-describedby="city-err">
        <option value="" selected disabled>Choose a city</option>
        <option value="lahore">Lahore</option>
        <option value="islamabad">Islamabad</option>
        <option value="rawalpindi">Rawalpindi</option>
        <option value="karachi">Karachi</option>
        <option value="faisalabad">Faisalabad</option>
        <option value="multan">Multan</option>
      </select>
      <span class="error-msg" id="city-err">Please choose a city.</span>
    </div>

    <div class="field">
      <label for="date" class="form-label">Delivery date</label>
      <input id="date" name="date" type="date" class="form-control"
             required min="2026-10-29" max="2026-11-02" aria-describedby="date-err">
      <span class="error-msg" id="date-err">Pick a date between 2026-10-29 and 2026-11-02.</span>
    </div>
  </fieldset>

  <div class="submit-row">
    <button type="submit" class="btn btn-lego btn-lg w-100">Place order</button>
  </div>
</form>
<!-- ===== END TASK 2 ===== -->
```

style.css – Task 2 CSS

```
/* ======================================================================
   TASK 2 - CHECKOUT FORM (N=4 -> BP=604px, H=28)
   ====================================================================== */

/* ---------- DEFAULT = BELOW BP: one column, labels above inputs----------*/
.checkout-form { display: grid; grid-template-columns: 1fr; gap: 1.5rem; max-width: 1100px; margin: 0
    auto; }

.checkout-fieldset {
  min-width: 0;                          /* fieldsets default to min-width:min-content, this stops
      overflow */
  border: 2px solid var(--lego-black);
  border-radius: 12px;
  padding: 1rem 1.25rem .25rem;
  background: #fff;
}
.checkout-fieldset legend { font-size: 1.15rem; font-weight: 800; color: var(--accent); }

.field { display: grid; grid-template-columns: minmax(0, 1fr); gap: .25rem; margin-bottom: 1rem; }
.field .form-label { margin: 0; font-weight: 700; }

/* error message hidden until the field is touched AND invalid */
.error-msg { display: none; color: #c00; font-size: .85rem; }

/* red border only AFTER interaction (:user-invalid), never on page load */
.form-control:user-invalid,
.form-select:user-invalid { border-color: #c00; box-shadow: 0 0 0 .15rem rgba(204,0,0,.2); }
.form-control:user-invalid + .error-msg,
.form-select:user-invalid + .error-msg { display: block; }

/* nice feedback when valid (accent colour) */
.form-control:user-valid,
.form-select:user-valid { border-color: var(--accent); }

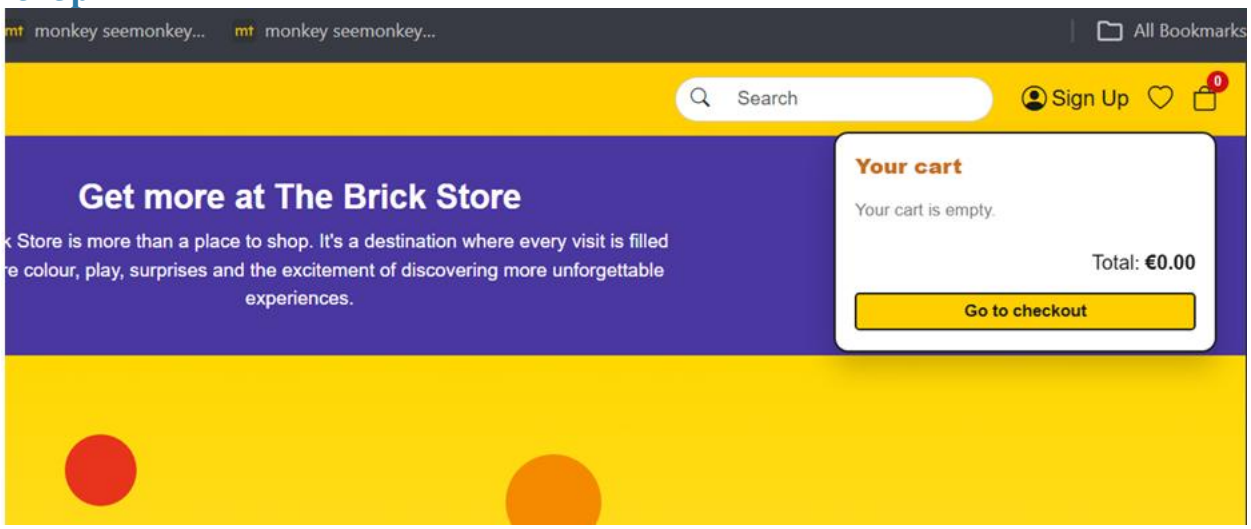
.submit-row { margin-bottom: 1rem; }

/* ---------- FROM BP AND WIDER: fieldsets side by side, labels in a left column----------*/
@media (min-width: 604px) {
  .checkout-form { grid-template-columns: 1fr 1fr; align-items: start; }
  .submit-row { grid-column: 1 / -1; }       /* button spans both columns */

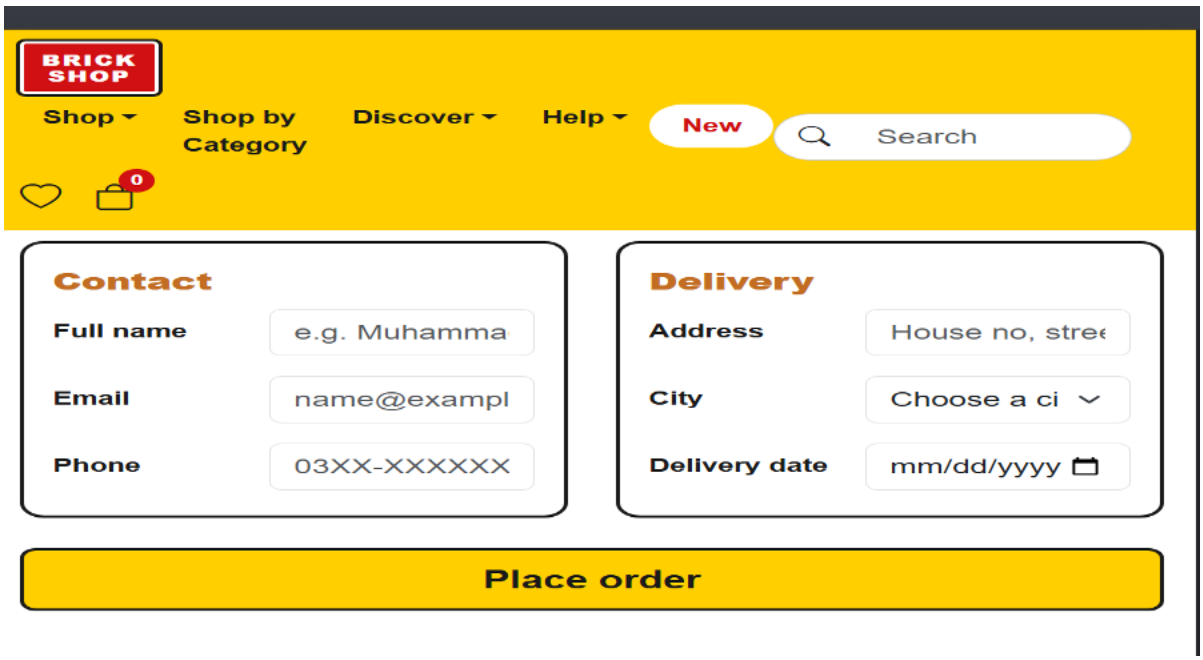
  .checkout-fieldset { padding: 1rem .9rem .25rem; }
  .field { grid-template-columns: 6rem minmax(0, 1fr); column-gap: .75rem; align-items: center; }
  .field .form-label { font-size: .9rem; }
  .field .error-msg { grid-column: 2; }      /* message sits under the input */
}
```

Task 3 – JavaScript cart badge and mini cart

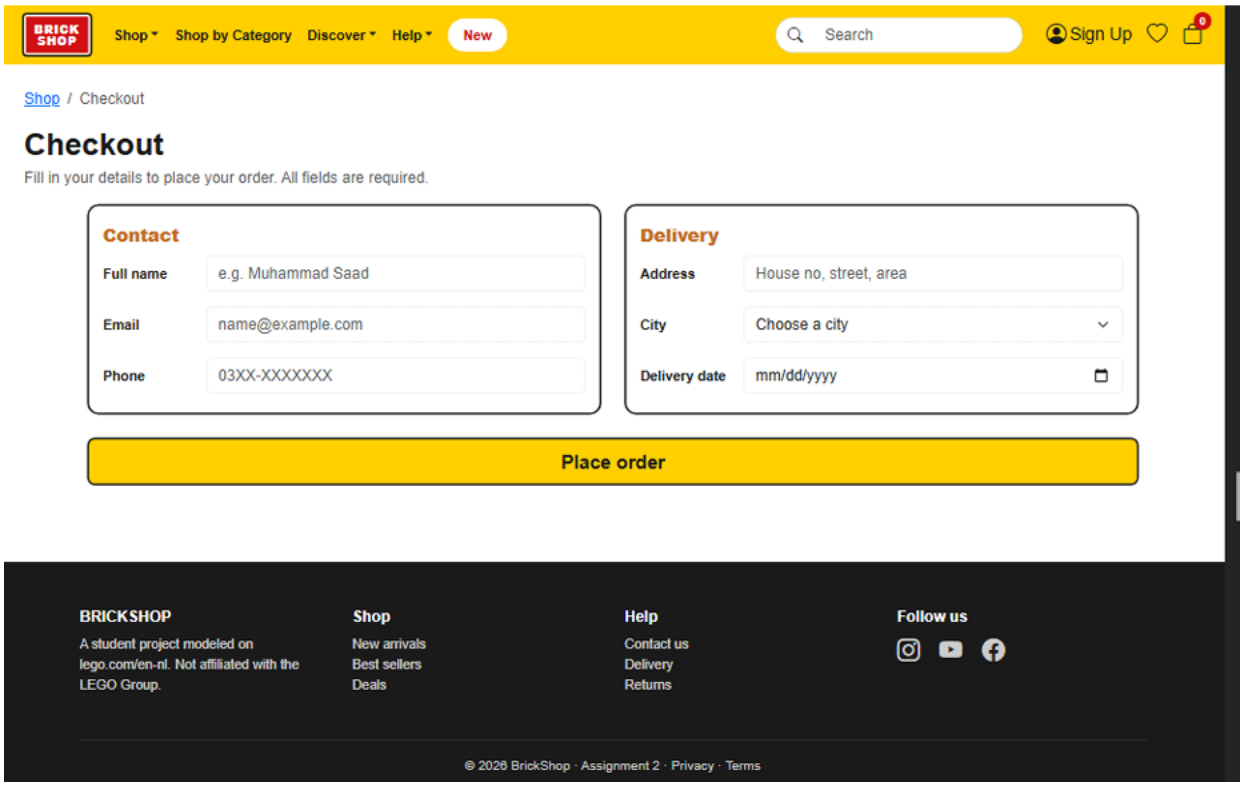
375px



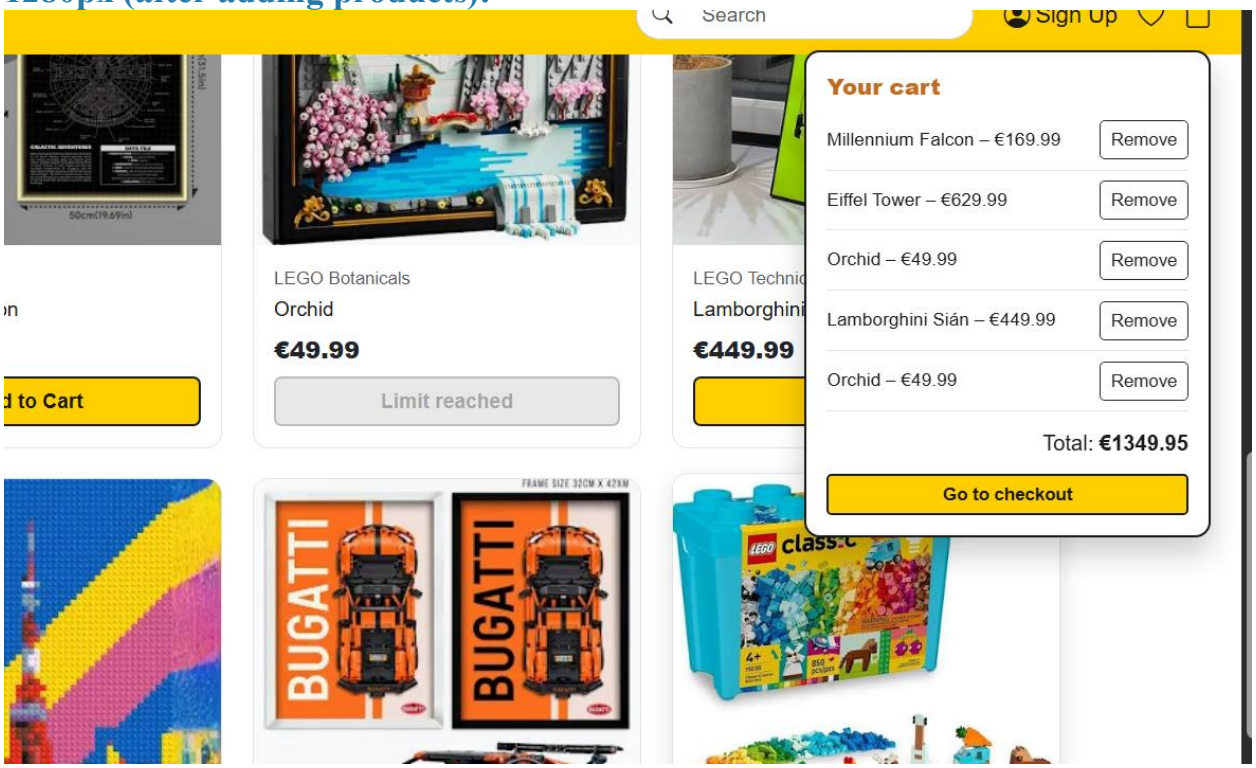
604px (BP)



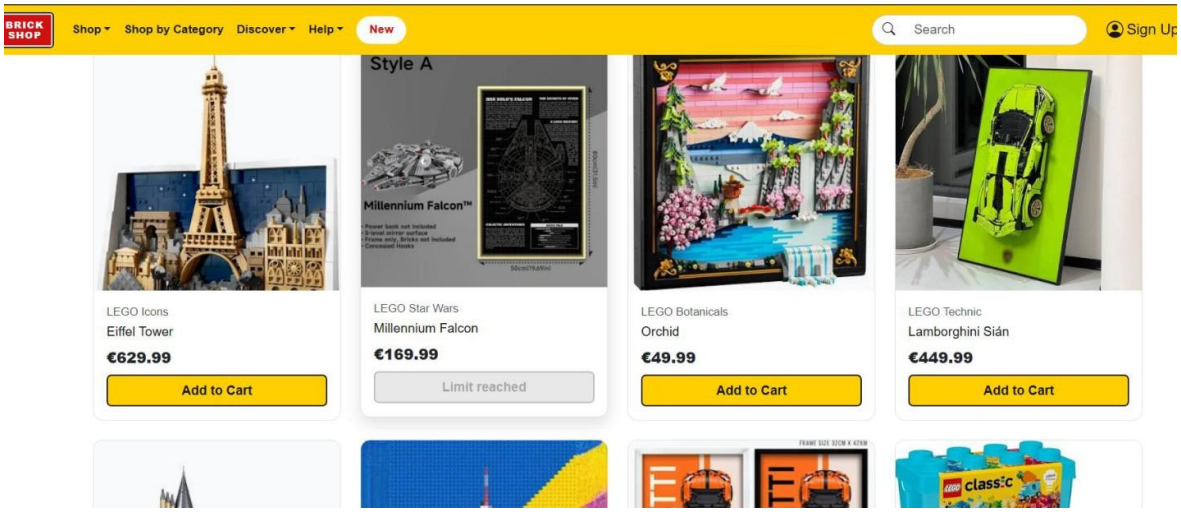
1280px (mini cart open, empty)



1280px (after adding products):



Limit reached (Millennium Falcon added 2 times, button disabled)



Task 3 – Code

script.js

```
/* ================================================================
   TASK 3 - Cart badge + mini cart  (plain JavaScript, no libraries)
   N = 4  ->  max copies of one product = (4 mod 4) + 2 = 2
   ================================================================ */

const MAX_PER_PRODUCT = 2;          // (N mod 4) + 2  -> change this if N changes

// ---- State: the cart is just an array of objects ----
let cart = [];                      // each item: { uid, name, price }
let nextUid = 1;                    // unique id so we can remove ONE specific row

// ---- DOM references (found once, reused) ----
const cartToggle = document.getElementById('cartToggle'); // the bag icon (link)
const miniCart = document.getElementById('miniCart'); // the panel
const cartList  = document.getElementById('cartList'); // <ul> inside the panel
const cartTotal = document.getElementById('cartTotal'); //  total price text
const cartBadge = document.getElementById('cartBadge'); // number on the icon
const addButtons = document.querySelectorAll('.product-card .btn-lego');

//  ----  Helpers  ----
function formatPrice(number) {
  return '€' + number.toFixed(2);
}

// Reads name and price from the card the button belongs to
function readProduct(button) {
  const card  = button.closest('.product-card');
  const name  = card.querySelector('.card-title').textContent.trim();
  const price = parseFloat(card.querySelector('.price').textContent.replace('€', ''));
  return { name, price };
}

function countOf(name) {
  return cart.filter(item => item.name === name).length;
}

// ---- Render everything from the cart array ----
function render() {
  // 1) badge
  cartBadge.textContent = cart.length;

  //  2)  list
  cartList.innerHTML = '';
  if (cart.length === 0) {
    const empty = document.createElement('li');
    empty.className = 'mini-cart-empty';
    empty.textContent = 'Your cart is empty.';
    cartList.appendChild(empty);
  }  else  {
    cart.forEach(item => {
      const li = document.createElement('li');
      li.className = 'mini-cart-item';

      const info = document.createElement('span');
      info.className = 'mini-cart-info';
      info.textContent = item.name + ' – ' + formatPrice(item.price);

      const remove = document.createElement('button');
      remove.type = 'button';
      remove.className = 'btn btn-sm btn-outline-dark remove-btn';
      remove.textContent = 'Remove';
      remove.dataset.uid = item.uid;

      li.appendChild(info);
      li.appendChild(remove);
      cartList.appendChild(li);
    });
  }

  // 3) total using reduce()
  const total = cart.reduce((sum, item) => sum + item.price, 0);
  cartTotal.textContent = formatPrice(total);
```

```
  // 4) buttons: disable when the limit is reached
  addButtons.forEach(button => {
    const { name } = readProduct(button);
    const limitReached = countOf(name) >= MAX_PER_PRODUCT;
    button.disabled = limitReached;
    button.textContent = limitReached ? 'Limit reached' : 'Add to Cart';
  });
}

// ---- Events ----

// Add to Cart (one listener per button)
addButtons.forEach(button => {
  button.addEventListener('click', () => {
    const product = readProduct(button);
    if (countOf(product.name) >= MAX_PER_PRODUCT) return; // safety check
    cart.push({ uid: nextUid++, name: product.name, price: product.price });
    render();
  });
});

// Remove (event delegation: ONE listener on the list handles all Remove buttons)
cartList.addEventListener('click', event => {
  const removeButton = event.target.closest('.remove-btn');
  if (!removeButton) return;
  const uid = Number(removeButton.dataset.uid);
  cart = cart.filter(item => item.uid !== uid);
  render();
});

// Cart icon opens / closes the mini cart (and stops the link from navigating)
cartToggle.addEventListener('click', event => {
  event.preventDefault();
  const isOpen = !miniCart.hidden;
  miniCart.hidden = isOpen;
  cartToggle.setAttribute('aria-expanded', String(!isOpen));
});

render();   // first draw: badge 0, empty message, buttons enabled
```

index.html – cart icon, badge and mini-cart panel

```
  <div class="cart-wrap">
    <a href="checkout.html" class="header-icon" id="cartToggle" aria-label="Shopping bag"
        aria-controls="miniCart" aria-expanded="false"><i class="bi bi-bag"></i><span class="cart-badge"
        id="cartBadge">0</span></a>

    <!-- ===== TASK 3: MINI CART PANEL ===== -->
     <div class="mini-cart" id="miniCart" hidden>
       <h2 class="mini-cart-title">Your cart</h2>
      <ul class="mini-cart-list" id="cartList"></ul>
      <p class="mini-cart-total">Total: <strong id="cartTotal">€0.00</strong></p>
      <a href="checkout.html" class="btn btn-lego btn-sm w-100">Go to checkout</a>
    </div>
  </div>
```

index.html – script link (before)

```
<script src="script.js"></script>
```

style.css – Task 3 CSS

```
/* ================================================================
   TASK 3 - CART BADGE + MINI CART    (cart limit per product = 2)
   ================================================================ */
.cart-wrap { position: relative; }                  /* reference box for the panel */
.cart-wrap > .header-icon { position: relative; }    /* reference box for the badge */

.cart-badge {
  position: absolute; top: -8px; right: -10px;
  min-width: 1.15rem; height: 1.15rem; padding: 0 .3rem;
  display: flex; align-items: center; justify-content: center;
  background: var(--lego-red); color: #fff;
  font-size: .7rem; font-weight: 700; border-radius: 999px;
}

.mini-cart {
  position: absolute; top: calc(100% + .75rem); right: 0;
```

```
  width: 320px; max-width: calc(100vw - 1.5rem);
  z-index: 1040;
  background: #fff; color: var(--lego-black);
  border: 2px solid var(--lego-black); border-radius: 12px;
  box-shadow: 0 12px 24px rgba(0,0,0,.2);
  padding: 1rem;
  font-size: 1rem;
}
.mini-cart-title { font-size: 1.1rem; font-weight: 800; margin: 0 0 .5rem; color: var(--accent); }
.mini-cart-list { list-style: none; margin: 0; padding: 0; max-height: 240px; overflow-y: auto; }
.mini-cart-item {
  display: flex; justify-content: space-between; align-items: center; gap: .5rem;
  padding: .5rem 0; border-bottom: 1px solid #e3e3e3; font-size: .9rem;
}
.mini-cart-empty { padding: .5rem 0; color: #777; font-size: .9rem; }
.mini-cart-total { margin: .75rem 0; text-align: right; font-size: 1.05rem; }
.btn-lego:disabled { background: #ddd; color: #777; border-color: #999; cursor: not-allowed; }
```